Dimension, Not Solvability: Trained Matrix States Converge to the Minimal Faithful Representation Dimension

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Abstract

When do differently structured tasks drive a learner to the same representation? We train identical matrix-state sequence models end-to-end on the word problems of five finite groups, S_3, S_4, A_5, S_5, A_6, spanning the solvable/non-solvable divide, and measure the effective rank of the learned state against each group's minimal faithful real representation dimension $d_{\min}$, the exact dimensionality representation theory assigns the task. The learned rank converges onto the algebraic minimum: group means 1.88/2.85/2.83/3.59/4.74 against $d_{\min} = 2/3/3/4/5$, all 19 seeds inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band, Spearman $\rho = 0.9747$, the maximum this design permits under the S_4/A_5 tie. The designed head-to-head is that tie: S_4 (solvable) and A_5 (non-solvable) share $d_{\min} = 3$, so if learned dimension were set by computational complexity class the pair should separate, and if set by representation dimension it should coincide. A pre-registered Welch equivalence test at ± 0.5 rank-units declares equivalence decisively (difference 0.019, both one-sided tests near seven times the critical value). A pre-registered causal force-rank razor adds a third leg: capping rank one below $d_{\min}$ pins recovery under 0.9 by the metric's own geometry ($\sqrt{(d_{\min}-1)/d_{\min}} < 0.9$ for every group tested), a floor no capped cell violates, and restoring $d_{\min}$ lifts it: recovery empirically clears the pre-registered $0.9\times$-anchor bar in every group. Learned representational dimension follows the task's algebra, not its complexity class.

1 Introduction

A central question for unifying accounts of neural representations is what, exactly, converges: when systems trained on different problems arrive at similar internal structure, which property of the problems predicts the shared solution? We study a setting where the answer can be computed in closed form before training. The word problem of a finite group G, mapping a sequence of generators to its composed product, admits an exact algebraic description of the minimal structure any faithful solution must carry: $d_{\min}(G)$, the dimension of G's smallest faithful real linear representation. The loss target already fixes $d_{\min}$ as the informative rank; the open question is whether the model's own recruited capacity, two ambient dimensions wider than $d_{\min}$, settles there too, tracking the group's algebra, or instead sorts with the group's computational class, solvable versus non-solvable, the divide at which word problems become NC^1-complete [Barrington, 1989] and at which state-tracking expressivity results separate recurrent architectures [Merrill et al., 2024, Grazzi et al., 2025].

We train one architecture, a Transformer-encoded single-matrix-state model under a hard bottleneck, on five groups spanning that divide, and measure the learned state's restricted effective rank against $d_{\min}$. Three results. First, convergence to the algebraic minimum: per-group mean rank lands within 11% of $d_{\min}$, all 19 seeds inside a pre-registered band, with Spearman $\rho = 0.9747$, the design's tie-capped maximum (Section 3). Second, the designed head-to-head at matched dimension: S_4 (solvable) and A_5 (non-solvable) share $d_{\min} = 3$, and a pre-registered equivalence test declares their learned ranks statistically equivalent, so the pair coincides on dimension rather than separating on complexity class (Section 4). Third, the converged dimension is causally load-bearing: capping rank at $d_{\min}-1$ pins recovery below the readout's own geometric ceiling in every group, and at $d_{\min}$ that ceiling lifts and recovery empirically returns (Section 5). A measurement lesson runs through all three: two instrument defects each produced a plausible wrong reading before being caught against raw run artifacts (Appendix B), and the convergence law above is what survived the repaired instrument.

2 Task, Model, and Instrument

Task. A word $w = g_{i_1} \cdots g_{i_L}$ is drawn as a random walk on G's Cayley graph over a symmetric generating set ($L \sim \mathrm{U}\{1, \ldots, 8\}$) and must map to $\rho_G(g_{i_1} \cdots g_{i_L})$, the composed product's image under a pinned orthogonal reference representation of dimension $d_{\min}(G)$. The five groups are S_3, S_4, A_5, S_5, A_6 with $d_{\min} = 2, 3, 3, 4, 5$ (realizations in Appendix A); S_3 and S_4 are solvable, the rest are not.

Model. A Transformer encoder with d_{state} learned row-reader queries writes each word into a single state $Z(w) \in \mathbb{R}^{d_{\text{state}} \times d_{\text{state}}}$, with $d_{\text{state}} = d_{\min}+2$ per group so that over-recruitment is expressible. The loss is cosine distance to the block-diagonal embedding of the reference ($\rho_G(\cdot) \oplus I_2$ in the observational sweep, $\rho_G(\cdot) \oplus 0$ in the causal wave); the readout is fixed, with no learned weights that could launder rank, and the decoder reads only Z, verified by a gradient blank-out test with a planted-leak negative control. Per-group step budgets (8k–40k, recorded in the design document's pre-registration) were pinned by convergence bars before the decisional sweep ran.

Instrument. The learned state is defined only up to scale and an orthogonal change of basis (a Schur intertwiner): $Z(w) \approx c\, Q\, [\rho_G(w) \oplus \cdot\,]\, Q^\top$. We estimate the model's own dominant $d_{\min}$-subspace U from the SVD of the *centered* covariance of $Z(w)$ over held-out words, measure **restricted effective rank**, the entropy effective rank of $U^\top Z(w) U$, and score **degauged recovery**, cosine to the reference after fitting (c, Q) on a fitting split and evaluating on a disjoint split, reported as $\mathrm{rec}@0.9$ (the fraction of eval words above cosine 0.9). Rank is invariant under the fitted gauge, so a rank-deficient state cannot be degauged into a full-rank one; centering is load-bearing (Appendix B). The observational band $[0.7, 1.3] \cdot d_{\min}$, the equivalence margin and sample size, the force-rank grid and its recovery bar, and every falsifier were fixed before the decisive runs.

3 Convergence to the Algebraic Minimum

Figure 1 is the headline: unconstrained trained states recruit restricted effective rank 1.877 ± 0.060, 2.852 ± 0.054, 2.832 ± 0.062, 3.591 ± 0.069, 4.736 ± 0.023 (mean $\pm$ sd over seeds) against $d_{\min} = 2, 3, 3, 4, 5$, within 11% of the algebraic minimum at every group (S_5 the largest deviation, 10.2%), with all 19 seeds inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Spearman correlation between learned rank and $d_{\min}$ is $\rho = 0.9747$, and this is the maximum the design can express: S_4 and A_5 tie at $d_{\min} = 3$ by construction, capping ρ below 1. Under the exact permutation null, $P(\rho \geq 0.8) = 8/120 \approx 6.7\%$. By pre-registration this observational leg is corroborating: the readout's $[1, d_{\min}]$ range makes the band's upper half non-binding by construction (Appendix B), and the dissociation is carried specifically by S_4/A_5 (Section 4), not the five-group trend; the causal weight is carried by Section 5. A disclosed length-robustness split moves per-group mean recovery cosine by at most 0.013 with no group-selective divergence (Appendix B).

4 Equivalence at Matched Dimension

The pair S_4/A_5 was designed into the family as the decisive contrast: the same $d_{\min} = 3$, opposite sides of the solvability divide, the axis on which expressivity theory separates state-tracking architectures [Merrill et al., 2024, Grazzi et al., 2025, Siems et al., 2025]. If learned representational dimension were set by complexity class, the pair should separate; if by representation dimension, it should coincide. Because the interesting outcome is a null, the comparison was pre-registered as an equivalence test rather than a difference test: Welch two-one-sided tests on restricted effective rank

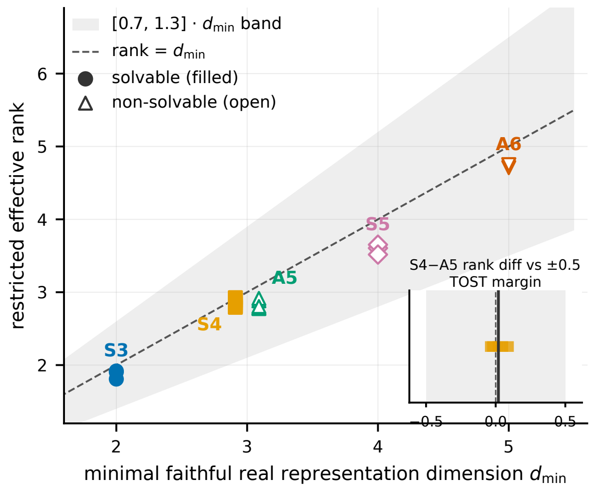

Figure 1: Learned rank converges to the minimal faithful representation dimension. Each point is one seed's restricted effective rank on the group word problem vs. its group's $d_{\min}$ (filled: solvable; open: non-solvable), with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band and the identity line; all 19 seeds are in band. Inset: per-seed $S_4 - \overline{A_5}$ rank differences at matched $d_{\min} = 3$ against the pre-registered ± 0.5 rank-unit equivalence margin (shaded); the vertical line is the observed mean difference, $+0.019$.

at margin ± 0.5 rank-units (half the spacing of the $d_{\min}$ ladder), $n = 5$ seeds per group, with a pre-run power simulation, fixed in the design record before the sweep ran, confirming the test reliably *rejects* equivalence at a true gap of 1.0 rank-units, so a real class effect of one ladder step could not have hidden inside the margin.

The observed difference is $+0.019$ rank-units (se 0.037, df 7.8), and both one-sided tests pass at roughly seven times the critical value ($t = 13.06$ and 14.12 against $t_{\text{crit}} = 1.865$): equivalence is declared decisively. The two groups' seed clusters are visually coincident in Figure 1 (inset). Convergence here is governed by what the algebra demands, not by how hard the class is to compute.

5 The Causal Check

Convergence and equivalence are observational; the third leg intervenes. The encoder's output is spectrally truncated to rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ throughout training against the zero-padded target $\rho_G(\cdot) \oplus 0$ of rank exactly $d_{\min}$, alongside an unconstrained anchor from the same family. Pre-registered reading: if $d_{\min}$ is load-bearing, $k = d_{\min}-1$ must fail and $k \geq d_{\min}$ must recover past $0.9\times$ the anchor's rec@0.9 on the pre-pinned conservative readout; a below-$d_{\min}$ cell clearing that bar would first indicate an instrument or embedding leak, not a live falsification of necessity, a contingency the geometric ceiling below closes outright, since no rank-($d_{\min}-1$) state can clear the bar without one.

Figure 2 shows a necessity floor at $d_{\min}$, not a plateau. The floor is set by the readout's own geometry, not discovered by training: $k = d_{\min}-1$ yields $\text{rec@0.9} = 0.000$ in all five groups and in all four independent seeds of the S_3 extension, exactly as the bound $\sqrt{(d_{\min}-1)/d_{\min}} < 0.9$ requires (Appendix B); the same cells' recovery cosines (0.61–0.84) sit under that ceiling, evidence the capped models reached their rank-constrained optimum rather than collapsing. Sufficiency is the empirical result: S_4, A_5, S_5, A_6 clear their pre-registered bars (0.800/0.700/0.600/0.650 against 0.585/0.630/0.450/0.585), in a majority of groups against a single noisy anchor seed rather than a stable ceiling, and S_3, whose decisive cell landed inside a pre-stated ± 0.05 marginality trigger, was extended to four seeds under the trigger's pre-registered routing: the seed-mean (0.5625) clears the fixed 0.495 bar, though only 2 of 4 seeds individually (Table 2). An earlier 58-cell sweep of this grid nulled through an instrument defect, an ambient identity block in the eye-padded target that taxed

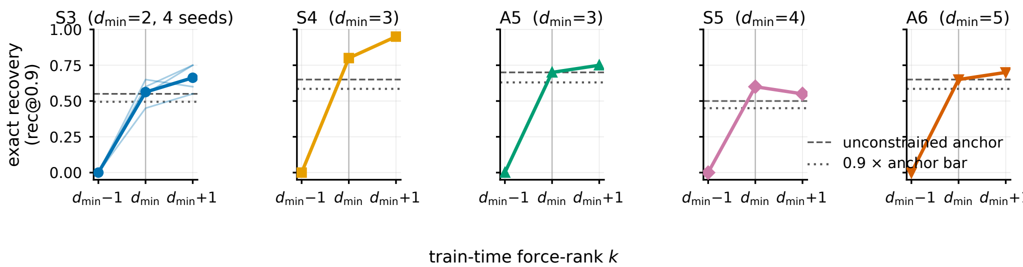

Figure 2: The converged dimension is causally load-bearing: per group, crosscheck $\mathrm{rec@0.9}$ on held-out words at train-time force-rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ (one seed per cell), vs. the unconstrained anchor (dashed) and the pre-registered $0.9\times$-anchor bar (dotted). Recovery is exactly 0.000 at $d_{\min}-1$ in every group, the readout's geometric floor (Appendix B), and returns at $d_{\min}$. S_3 overlays its four seeds (thin lines; bold mean); its below-$d_{\min}$ zero is unanimous.

every rank-capped arm; the verdict was registered as inconclusive with the tax established from the raw artifacts, and the zero-padded wave above is the registered fix (Appendix B).

6 Related Work and Limitations

Merrill et al. [2024], Grazzi et al. [2025], and Siems et al. [2025] frame state tracking by what recurrent architectures *can* express on exactly these word problems, the solvability axis; this work measures what trained states *do* converge to, and finds the learned dimension sorts by representation theory instead. Nazari and Rusch [2026] and Sun et al. [2026] measure state-rank dynamics in pretrained linear-attention models observationally, with no algebraic ground truth to converge to and no causal leg. Nichani et al. [2025] prove associative-memory capacity under argmax decoding, the loophole our exact-continuous-recovery scoring closes; Mishra et al. [2026] train matrix-state RNNs on S_3 composition but measure no state rank and run no rank intervention. Chughtai et al. [2023] reverse-engineer which representation-theoretic structure networks recruit on finite-group composition tasks; this work asks a narrower question on the same substrate, at what dimension, with a pre-registered causal intervention absent there. Huh et al. [2024] propose that models trained on different tasks converge to a shared representation; the present law is a narrower, algebra-driven instance of that convergence.

Limitations. All results are on sub-1M-parameter synthetic models under a deliberate single-state bottleneck; nothing here is a claim about pretrained networks, and the family is five groups with $d_{\min} \in \{2, \ldots, 5\}$ at $d_{\text{state}} = d_{\min} + 2$. Causal cells are single-seed except S_3 (four seeds), with the necessity zeros exact wherever seeds exist; two groups at the shortest pinned budget (S_3, S_5) show soft convergence and are also the two with the largest observational deviation and the one marginal causal cell, a pre-registered rather than post hoc budget schedule but a caveat on cross-group comparison strength. The observational readout partially favors rank $\approx d_{\min}$ by construction, which is why it was pre-registered as corroborating and the causal wave carries the claim.

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A The Five Groups and Reference Representations

G	$\lvert G\rvert$	$d_{\min}$	solvable	realization of ρ_G
S_3	6	2	yes	standard (dihedral)
S_4	24	3	yes	cube rotations
A_5	60	3	no	icosahedral rotations
S_5	120	4	no	standard (zero-sum)
A_6	360	5	no	standard (zero-sum)

Table 1: The five task groups. $d_{\min}$ is the minimal faithful real representation dimension; each reference representation is orthogonal by construction and verified faithful (injective on G) at build time. S_4/A_5 form the designed pair: matched $d_{\min}$, opposite solvability.

B Instrument Details and the Two Defects

Centering. For an orthogonal target the uncentered covariance of $Z(w)$ is isotropic and carries no subspace information; the production lens centers it (a nontrivial irreducible representation has zero group mean, cancelling the constant block). The step is load-bearing: an uncentered lens scores a flawless synthetic model 0.705 where the centered lens scores 0.9996 on identical data.

Length-robustness split. The disclosed split referenced in Section 3 scores only words of length $L \geq 2$ (the $L = 1$ read is attention-degenerate and was demoted with a mechanism note in the run records): it moves per-group mean recovery cosine by at most 0.013 (per-seed at most 0.041), with no group-selective divergence.

The ambient-identity tax. The first 58-cell sweep's force-rank target was the eye-padded $\rho_G(\cdot)\oplus I_2$, whose rank is $d_{\text{state}} = d_{\min} + 2$ for every group. The constant identity block is the cheapest reliable loss reduction, so a rank-capped arm buys it first, leaving effective budget $k - 2 < d_{\min}$ for ρ_G at every grid point; no capped cell in that sweep ever tested the confirm direction. The raw-artifact signature: the 39 force-rank cells' direct cosine matched the rank-k optimum $\sqrt{k/d_{\text{state}}}$ with mean absolute deviation 0.028 (maximum 0.166; the two S_5 below-$d_{\min}$ cells are the only outliers, a disclosed optimization shortfall), so the arms had trained to their rank-constrained ceiling rather than under-training. The sweep verdict was registered as inconclusive with the tax as mechanism;

the zero-padded causal wave of Section 5 is the registered fix, its 30 cells configuration-verified one-by-one against a manifest re-derived independently from the design record. In that wave an eye-padded corroboration arm reproduces the failure on demand at raw $k = d_{\min}+1$ (effective budget $d_{\min}-1$) while the tax-paid point recovers.

The observational band's non-binding upper half. `entity_subspace_from_words` returns the top-$d_{\min}$ left singular vectors of the centered covariance by construction, so the restricted state $U^\top ZU$ is always $d_{\min} \times d_{\min}$, and the entropy effective rank of a $d_{\min} \times d_{\min}$ input has range $[1, d_{\min}]$. Restricted effective rank therefore cannot exceed $d_{\min}$, and so cannot exceed $1.3 \cdot d_{\min}$, for any seed of any group; only the band's lower half, $\geq 0.7 \cdot d_{\min}$, is a test the data could fail. Every group mean lands in $[0.895, 0.951] \cdot d_{\min}$.

The rank-constrained cosine ceiling. The causal wave's target is $\rho_G(\cdot) \oplus 0$, of rank exactly $d_{\min}$ with all $d_{\min}$ informative singular values equal to 1 (the reference representation is orthogonal by construction, Appendix A). For any matrix M of rank $\leq k \leq d_{\min}$, von Neumann's trace inequality gives $\langle M, T\rangle_F \leq \sum_{i=1}^{k} \sigma_i(M)\, \sigma_i(T) = \sum_{i=1}^{k} \sigma_i(M)$, and Cauchy–Schwarz bounds $\sum_{i=1}^{k} \sigma_i(M) \leq \sqrt{k}\, \|M\|_F$, so $\cos(M, T) \leq \sqrt{k/d_{\min}}$. At $k = d_{\min}-1$ this ceiling is $0.707/0.816/0.816/0.866/0.894$ for $S_3/S_4/A_5/S_5/A_6$, below the $\mathrm{rec@0.9}$ threshold of 0.9 in every case: no rank-$(d_{\min}-1)$ state, however trained, can register a single word above cosine 0.9 against this target. The necessity result ($\mathrm{rec@0.9} = 0.000$ at $d_{\min}-1$, all groups, all seeds) is this bound realized, not an independent discovery about what SGD finds; the corresponding empirical content is that the observed cosines at $d_{\min}-1$ (0.61–0.84) sit under their respective rank-constrained optima, evidence optimization reaches the geometry's own ceiling. At $k \geq d_{\min}$ the ceiling is 1.0 (exact recovery is representable), so recovery above 0 at $d_{\min}$ is not guaranteed by the metric and carries the causal weight (sufficiency).

Primary and crosscheck degauging. The design record's default pipeline treats scale-only degauging ($\hat{Q} = I$) as the primary metric, with the fitted-(c, Q) Procrustes score retained as a robustness cross-check, after calibration on unconstrained checkpoints found $\hat{Q} \approx I$ empirically. Under the force-rank grid's zero-padded target this equivalence breaks: the informative block's degenerate singular spectrum makes the scale-only fit's basis arbitrary, so the pre-registration for the causal wave specifically pins the fitted-(c, Q) score, denoted $\mathrm{rec@0.9}$ throughout Section 5 and Figure 2, as decisional; the scale-only score informs no reported conclusion. The two diverge sharply on this grid, which is why the distinction is disclosed here.

C S_3 Per-Seed Causal Detail

seed	anchor	$k{=}d_{\min}{-}1$	$k{=}d_{\min}$	$k{=}d_{\min}{+}1$	own bar	clears
0	0.550	0.000	0.450	0.550	0.495	no (-0.045)
1	0.600	0.000	0.550	0.750	0.540	yes ($+0.010$)
2	0.800	0.000	0.600	0.750	0.720	no (-0.120)
3	0.600	0.000	0.650	0.600	0.540	yes ($+0.110$)

Table 2: S_3 four-seed causal detail (crosscheck $\mathrm{rec@0.9}$) behind the seed-mean confirmation in Section 5; the own bar is $0.9\times$ each seed's anchor. $k = d_{\min}-1$ reads exactly 0.000 in all four seeds; $k = d_{\min}$ clears each seed's own bar in 2 of 4 (seeds 1, 3), missing in the other 2 because the anchor itself ranges 0.550–0.800 across seeds. The pre-registered criterion compares the seed-mean $k = d_{\min}$ value (0.5625) against the bar fixed before the extension ran (0.495, from seed 0's anchor), specifically to avoid computing the bar from the same noisy seeds it is judged against; recomputing the bar as $0.9\times$ the four-seed anchor mean (raw mean 0.6375, bar 0.574) would put the seed-mean value 0.011 below it.

D Per-Seed Observational Rank

G	$d_{\min}$	n	restricted eff. rank	band	in band
S_3	2	3	1.877 ± 0.060	$[1.4, 2.6]$	3/3
S_4	3	5	2.852 ± 0.054	$[2.1, 3.9]$	5/5
A_5	3	5	2.832 ± 0.062	$[2.1, 3.9]$	5/5
S_5	4	3	3.591 ± 0.069	$[2.8, 5.2]$	3/3
A_6	5	3	4.736 ± 0.023	$[3.5, 6.5]$	3/3

Table 3: Restricted effective rank of the unconstrained trained state (mean $\pm$ sd over seeds) against each group's $d_{\min}$, with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Every seed of every group is in band; Spearman $\rho = 0.9747$ is the tie-capped maximum; the dimension-matched pair S_4/A_5 differs by 0.019 rank-units.